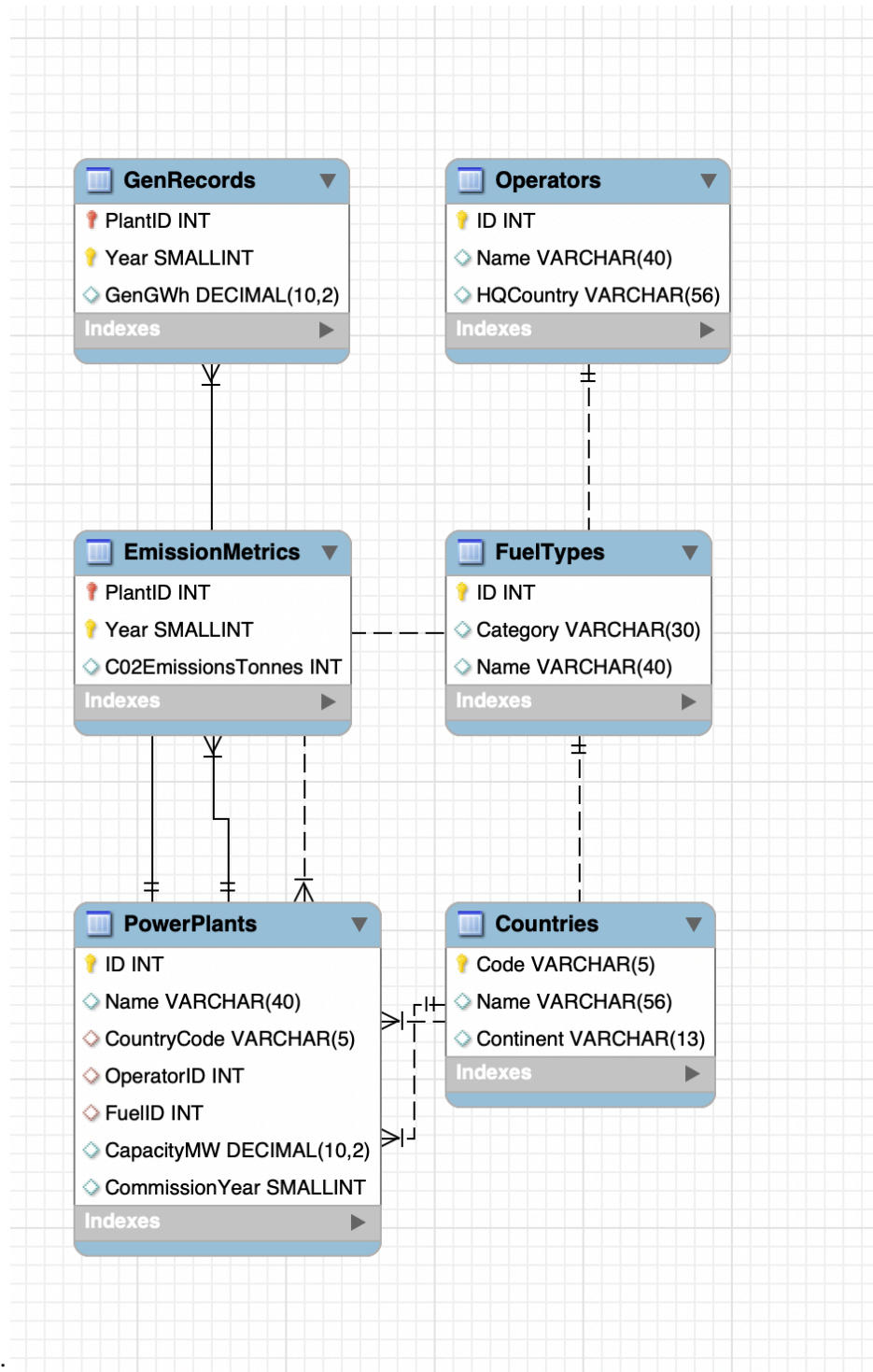




EER Diagram:


```
4
-- write 4
with total_cum_power_gen as (
  select operators.operatorname, HQCountry, sum(Generation) as Gen
  from operators
  left join power_plants on power_plants.operatorid = operators.operatorid
  left join generation_records on generation_records.powerplantid = power_plants.powerplantid
  group by operators.operatorname, operators.HQCountry
)
select * from total_cum_power_gen
order by gen desc; -- only seven rows because one Operator isn't in power_plants
```

4.sql

operatorname	HQCountry	Gen
China Yangtze Power	CHN	169800.30
Electrobras	BRA	89000.10
NTPC Limited	IND	35600.80
Global Hydro Inc	USA	21500.50
Rhein Power AG	DEU	18400.20
Snowy Hydro	AUS	14200.40
Tepco Energy	JPN	0.00
Eskom Holdings	ZAF	NULL

```
5
-- write 5
with total_gen as (
  select countries.countrycode, sum(Generation) as Generation
  from countries
  join power_plants on countries.countrycode = power_plants.countrycode
  join generation_records on generation_records.powerplantid = power_plants.powerplantid
  group by countries.countrycode
),
total_emissions as (
  select countries.countrycode, sum(CO2EmissionsTonnes) as Emissions
  from countries
  join power_plants on countries.countrycode = power_plants.countrycode
  join emission_metrics on emission_metrics.powerplantid = power_plants.powerplantid
  group by countries.countrycode
)
select countries.countryname, total_gen.generation, total_emissions.emissions
from countries
join total_emissions on total_emissions.countrycode = countries.countrycode
join total_gen on total_gen.countrycode = countries.countrycode
order by generation desc;
```

5.sql

countryname	generation	emissions
China	169800.30	5200000
Brazil	89000.10	0
India	35600.80	3100000
United States	21500.50	0
Germany	18400.20	2450000
Australia	14200.40	1950000
Japan	0.00	0